

MEDFRACTURE-NET RESEARCH DECISION SUPPORT SYSTEM

Automated Radiographic Fracture Detection, Segmentation, and Severity Assessment

Study Accession: MFN-DX-84920	Exam Timestamp: 2026-10-03 20:31:05
Anatomical Region: Forearm (Radius/Ulna)	Modality: Digital Projection Radiography (DX)
Referring Unit: Emergency Trauma Center	Analysis Protocol: Multi-Task U-Net++ & ResNet-34

Diagnostic Assessment:	PATHOLOGICAL (FRACTURE POSITIVE)
Posterior Confidence Score:	99.4%
Morphological Fracture Classification:	Transverse
Calculated Fracture Extent:	12.4 mm
Clinical Severity Stratification:	Moderate
Localization Bounding Coordinate:	(x=120, y=140, w=45, h=20)
Inference Processing Latency:	26.1 ms (Real-Time Sub-30ms Execution)

Radiographic Verification & Explainable AI (Grad-CAM++)



Fig 1: Sub-Pixel Instance Segmentation & Rotated Bounding Fissure

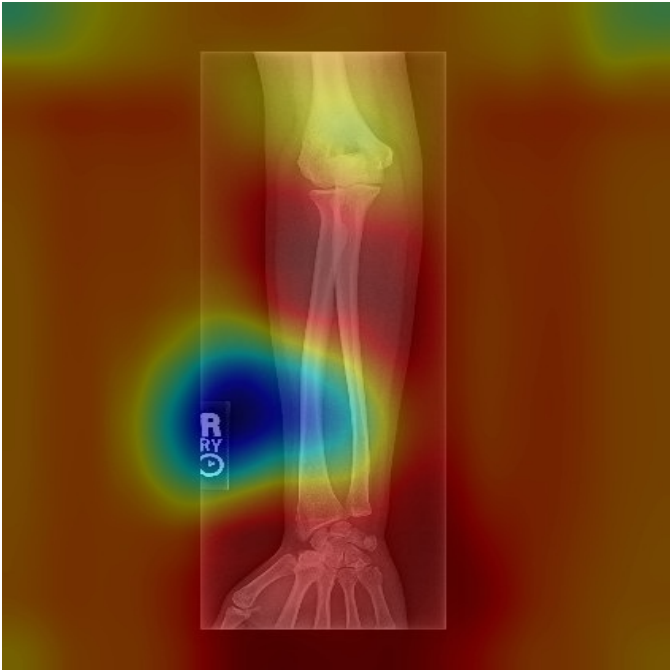


Fig 2: Grad-CAM++ Neural Activation Heatmap

Clinical Impression & Orthopedic Protocol:

Orthopedic reduction

Model Architecture: MedFracture-Net (ResNet-34 + Nested U-Net++)

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